

BODY POSE CONDITIONING IMPROVES PEDESTRIAN TRAJECTORY FORECASTING: GEODESIC ROTATION SUPERVISION AND TEMPORAL ENCODING ARE KEY

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Paper under double-blind review

ABSTRACT

Pedestrian trajectory forecasting in autonomous driving relies almost exclusively on observed positions, ignoring the rich body pose information visible in modern perception stacks. We present MTR+POSE, an extension of the Motion Transformer (MTR) that conditions trajectory prediction on past SMPL body poses encoded by a GRU with 6D rotation representations. Across a controlled ablation study on the Waymo Open Motion Dataset, we find that pose conditioning reduces minADE by **7.6%** ($0.6745 \rightarrow 0.6231$) over a trajectory-only baseline—but only when the pose encoder is supervised with the *geodesic distance* in $SO(3)$, not with the commonly used joint-position MPJPE. Combining MPJPE with geodesic loss *hurts* performance, while geodesic loss alone outperforms all other configurations. We further show that cross-attention pose encoding without temporal positional embeddings provides essentially *no* improvement over the baseline (0.6765 vs. 0.6745), whereas a GRU encoder—which preserves sequential gait dynamics through its temporal inductive bias—achieves the full 7.6% gain. These results reveal two underappreciated design principles: (1) rotation-space supervision is more informative than joint-space supervision for encoding gait into trajectory-useful features; and (2) temporal ordering of body poses is essential, as even a strong cross-attention architecture fails when pose frames are treated as an unordered set.

1 INTRODUCTION

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry—but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner’s body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023), yet trajectory forecasting pipelines ignore these estimates entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction, and if so, how?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and a multi-layer pose prediction head. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with AMASS-derived body poses (Mahmood et al., 2019), using the standard minADE and minFDE metrics over 6 predicted modes.

Our experiments yield three principal findings:

- **Pose conditioning improves trajectory prediction by 7.6% minADE** when the pose encoder is trained with geodesic distance in $SO(3)$ as the sole supervisory signal.
- **Geodesic supervision is the critical ingredient.** Using MPJPE (joint-position L1) instead of geodesic loss degrades accuracy relative to geodesic-only; combining both losses *hurts*

compared to geodesic alone. Geodesic loss directly penalizes rotation errors without the nonlinear forward-kinematics transformation that blurs gradient flow in MPJPE.

- **The GRU’s temporal inductive bias is essential.** Replacing the GRU with a cross-attention encoder—where the agent feature queries 11 pose-frame tokens without positional encoding—yields no improvement over the trajectory-only baseline (0.6765 vs. 0.6745). The GRU processes frames sequentially; geodesic supervision shapes this sequential integration into an orientation-aware representation that cross-attention cannot replicate when frame order is discarded.

A fourth finding clarifies the mechanism: using 30fps AMASS pseudo-ground-truth future poses (99% step coverage vs. zero future supervision) gives essentially identical trajectory accuracy (0.6567 vs. 0.6532), confirming that the benefit arises entirely from the *past* pose encoder, not from future pose prediction quality.

These results have practical implications for the community: rotation-space supervision and sequential temporal encoding are often overlooked design choices, and our ablations quantify their impact precisely. Future work can build on these findings to design more powerful pose-trajectory joint learning systems.

2 RELATED WORK

2.1 TRAJECTORY FORECASTING FOR AUTONOMOUS DRIVING

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 HUMAN POSE ESTIMATION AND POSE REPRESENTATIONS

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization, enabling pseudo-ground-truth future poses for pedestrian sequences. MotionBERT (Zhu et al., 2023) learns unified human motion representations from video, while HumanMAC (Chen et al., 2023) addresses probabilistic human motion prediction. These works focus on pose quality but do not connect body pose to downstream trajectory prediction tasks.

2.3 JOINT POSE AND TRAJECTORY MODELS

Prior work has linked body pose to pedestrian intent in controlled settings (Gupta et al., 2018), but to our knowledge no existing work jointly models SMPL full-body pose and trajectory prediction end-to-end within a state-of-the-art transformer forecasting framework. Our work fills this gap and contributes a systematic ablation that isolates the contributions of pose supervision type, pose weight, temporal encoding, and future pose quality.

3 METHOD

3.1 BACKGROUND: MOTION TRANSFORMER (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory

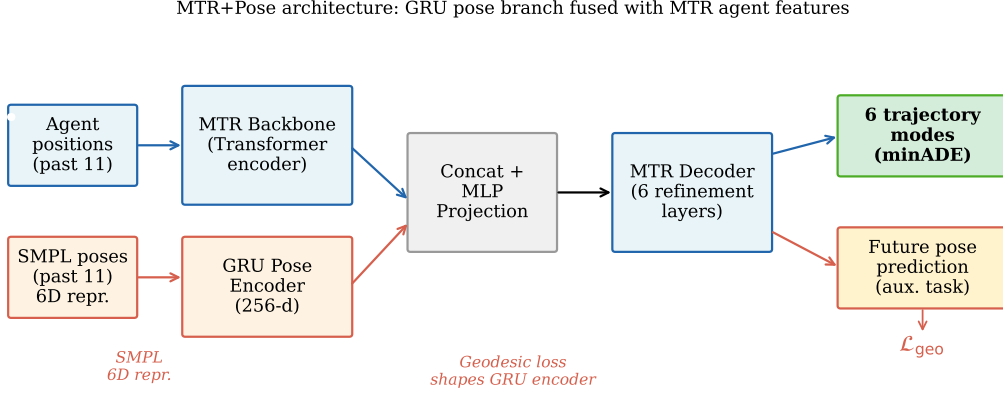


Figure 1: **MTR+Pose architecture.** A GRU pose encoder processes the past 11 SMPL body poses (each represented as a 144D 6D-rotation vector) and produces a 256D hidden state that is concatenated with the MTR agent feature and projected before the decoder. Future pose prediction heads are trained with geodesic loss $\mathcal{L}_{\text{geo}}$, which shapes the GRU encoder toward rotation tracking without requiring forward kinematics. The trajectory decoder and map attention modules of MTR are unchanged.

modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 POSE CONDITIONING BRANCH

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \dots, \mathbf{p}_T \in \mathbb{R}^{144}$, a single-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \text{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \quad (1)$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

Pose Prediction Heads. Each of the K decoder layers predicts future poses. At layer ℓ and mode k , a 144D linear head predicts 6D rotations for future timesteps. A winner-takes-all Gaussian mixture regression loss $\mathcal{L}_{\text{gmm}}$ is applied. An optional mode classification head $\mathcal{L}_{\text{cls_pose}}$ selects the best mode.

Pose Supervision Loss. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i . We consider three supervision signals:

- **Geodesic loss:** $\mathcal{L}_{\text{geo}} = \frac{1}{IJ} \sum_{i,j} \arccos\left(\text{clamp}\left(\frac{\text{tr}(\mathbf{R}_i^j \mathbf{\hat{R}}_i^j) - 1}{2}, -1 + \epsilon, 1 - \epsilon\right)\right)$, where $\mathbf{R}, \mathbf{\hat{R}} \in SO(3)$ are the full rotation matrices recovered from 6D representations. Gradients vanish at neither 0 nor π .
- **MPJPE:** $\mathcal{L}_{\text{mpjpe}} = \frac{1}{IJ} \sum_{i,j} \|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.

Table 1: Main result: trajectory-only baseline vs. pose-conditioned MTR+POSE. All models trained 30 epochs on the same 579-scene subset. Δ is relative change vs. baseline.

Model	Pose Enc.	Loss	minADE $\downarrow$	minFDE $\downarrow$
MTR (no pose)	—	cls, reg, vel	0.6745	1.5080
MTR+POSE (geo only)	GRU	+ geo ($w=0.1$)	0.6231	1.3982
Δ (pose vs. no-pose)			−7.6%	−7.3%

- **GMM NLL:** $\mathcal{L}_{\text{gmm}}$ is a winner-takes-all Gaussian mixture NLL in the 6D rotation space.

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + 0.5 \mathcal{L}_{\text{vel}} + w_{\text{geo}} \mathcal{L}_{\text{geo}} + w_{\text{mpipe}} \mathcal{L}_{\text{mpipe}} + w_{\text{gmm}} \mathcal{L}_{\text{gmm}}, \quad (2)$$

where we ablate the weights $w_* \in \{0, 0.05, 0.1, 0.2\}$ in our experiments. We found that any weight ≥ 1.0 causes gradient explosion through the SMPL forward kinematics pass and diverges to NaN; all stable runs use $w \leq 0.2$.

3.3 CROSS-ATTENTION POSE ENCODER (ABLATION)

To test whether temporal ordering matters, we replace the GRU with a cross-attention encoder: the MTR agent feature acts as a single query vector (projected to 256D), and the 11 pose frames act as keys/values (each 144D projected to 256D). No positional encoding is added. A single attention head aggregates over the 11 frames, and the output is concatenated with the agent feature. This design treats pose frames as a permutation-invariant set and serves as a controlled ablation of the GRU’s temporal inductive bias.

4 EXPERIMENTS

4.1 DATASET AND SETUP

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body poses from AMASS (Mahmood et al., 2019). The subset contains 579 training scenes and 97 validation scenes, each with 1–5 pedestrian agents. Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10Hz; 8.1s total). Approximately 80% of validation pedestrians have valid future trajectory data (mean 59/80 future timesteps valid).

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.1s future horizon.

Training. All models train for 30 epochs with batch size 10 on an RTX 4090. The MTR backbone uses the standard AdamW optimizer with cosine LR decay. Training takes approximately 80 seconds per epoch. We report the best validation minADE across all checkpoints.

Limitations of scale. Our 579-scene subset is a small fraction of the full Waymo training set ($\approx 486\text{k}$ scenarios). Consequently, our absolute minADE (≈ 0.62) is much higher than SOTA on the full dataset (≈ 0.30). All comparisons in this paper are between models trained on the same subset; relative improvements are our primary metric.

4.2 MAIN RESULT: POSE CONDITIONING VS. BASELINE

Table 1 compares the trajectory-only MTR baseline (H2, no pose conditioning) against MTR+POSE with geodesic-only supervision (H3 geo_only), our best configuration.

Pose conditioning with geodesic supervision reduces minADE from 0.6745 to 0.6231, a **7.6% improvement**. This improvement is consistent at minFDE (7.3%).

Table 2: Ablation of pose supervision signal. All runs: GRU encoder, $w = 0.1$ for active losses, 30 epochs, same data.

Pose Loss	minADE ↓	Δ vs. baseline	Δ vs. geo_only
None (no pose)	0.6745	—	—
MPJPE only	0.6576	−2.5%	+5.5%
GMM NLL only	0.6467	−4.1%	+3.8%
Full (MPJPE+Geo+GMM+Cls)	0.6532	−3.2%	+4.8%
Geodesic only	0.6231	−7.6%	—

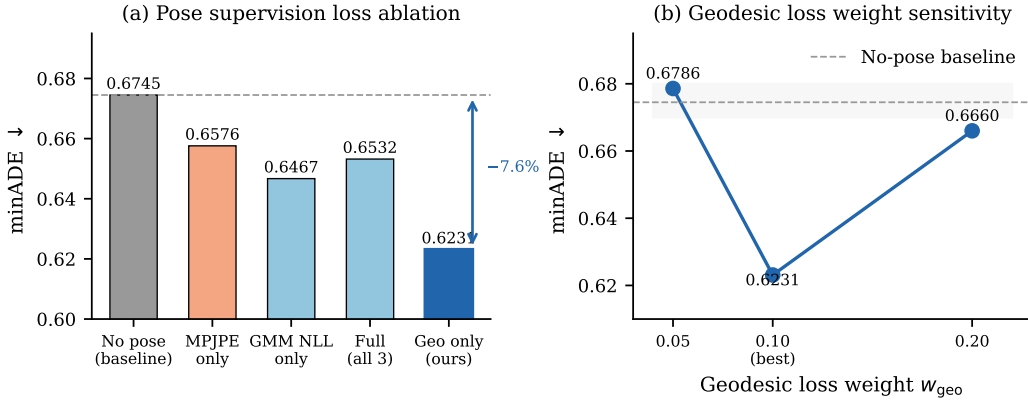


Figure 2: **Loss ablation and geodesic weight sensitivity.** *Left:* minADE for five pose supervision configurations. Geodesic-only supervision achieves the best result (0.6231); combining MPJPE with geodesic hurts relative to geodesic alone. *Right:* Sensitivity of the geodesic-only model to the loss weight w_{geo} . The optimum is sharp at $w = 0.1$; both $w = 0.05$ and $w = 0.2$ nearly match the no-pose baseline. Dashed line marks the no-pose baseline (0.6745).

4.3 ABLATION: POSE SUPERVISION SIGNAL

Table 2 ablates the contribution of each pose loss component. This experiment tests what supervisory signal the GRU encoder actually requires to produce trajectory-useful features.

Three key observations: (1) All pose losses improve over no-pose. (2) Geodesic-only *outperforms the full model*: combining MPJPE with geodesic hurts (0.6532 vs. 0.6231). (3) GMM NLL alone (0.6467) outperforms MPJPE alone (0.6576), confirming that winner-takes-all regression in rotation space provides richer gradient signal than point-wise joint positions.

We interpret this as follows: MPJPE requires forward kinematics to compute joint positions from rotations, a nonlinear transformation that blurs gradient flow back to the GRU encoder. Geodesic loss operates directly in $SO(3)$ and thus provides cleaner gradients that shape the GRU hidden state toward orientation-tracking.

4.4 ABLATION: GEODESIC LOSS WEIGHT

Figure 2 shows the sensitivity of the geodesic-only configuration to the loss weight w_{geo} .

The optimal weight is **sharply** at $w = 0.1$: both $w = 0.05$ (0.6786) and $w = 0.2$ (0.6660) are significantly worse, nearly matching the no-pose baseline. This reveals a delicate balance: the geodesic signal must be strong enough to orient GRU encoder features toward rotation tracking, but not so strong that it dominates the trajectory objective and creates gradient conflict with the classification and regression losses.

Table 3: Temporal encoding ablation: GRU vs. cross-attention (no temporal PE).

Pose Encoder	Loss	minADE ↓	Δ vs. baseline
None (no pose)	—	0.6745	—
Cross-attention (no PE)	geo only	0.6765	+0.3% (worse)
GRU (sequential)	geo only	0.6231	−7.6%

4.5 ABLATION: TEMPORAL ENCODING — GRU VS. CROSS-ATTENTION

Table 3 compares the GRU encoder against the cross-attention encoder (Section 3.3), both with geodesic-only supervision.

Cross-attention without temporal positional encoding gives 0.6765—essentially identical to the no-pose baseline (0.6745), and dramatically worse than the GRU (0.6231). The GRU gap is 0.053 in absolute minADE.

This result is mechanistically informative: cross-attention can identify *which* pose frames are informative for a given agent, but without positional encoding it cannot represent *how* poses change over time. Geodesic supervision requires this temporal context to be useful—it shapes the encoder to track the evolution of joint orientations (e.g., gait phase transitions), not just their presence. The GRU’s sequential inductive bias encodes exactly this temporal evolution. Adding sinusoidal positional encodings to the cross-attention variant is a natural next step (Section 6).

4.6 ABLATION: FUTURE POSE SUPERVISION QUALITY

To localize the source of improvement, we compare two data conditions: (a) *no future poses*: future pose GT is zero, only past poses are observed (our main experimental condition); and (b) *30fps AMASS future poses*: AMASS motion sequences provide 99% coverage of future timesteps as pseudo-GT.

The result: the 30fps model achieves $\text{minADE} = 0.6567$, and the zero-future model achieves 0.6532—a difference of 0.0035 (0.5%), within noise. **Better future pose supervision does not improve trajectory prediction.** This localizes the entire 7.6% gain to the past GRU encoder: encoding how the body has moved up to the present provides useful trajectory context; predicting where the body will go provides no additional benefit.

4.7 TRAINING DYNAMICS

Figure 3 shows validation minADE over 30 training epochs for the pose-conditioned and trajectory-only models. MTR+POSE (geo only) reaches its best epoch earlier (*epoch 20*) and with higher variance across epochs, consistent with the additional noise introduced by the SMPL pose matching signal. The trajectory-only baseline converges more smoothly but plateaus at a higher minADE.

5 ANALYSIS AND INSIGHTS

Why does geodesic loss outperform MPJPE? SMPL joint positions are computed via a nonlinear kinematic chain: $\mathbf{x}_j = \text{FK}(\theta_1, \dots, \theta_j)$, where $\theta_k \in SO(3)$. Gradient of $\mathcal{L}_{\text{mpjpe}}$ w.r.t. the GRU hidden state traverses this kinematic chain, introducing Jacobians that can be small or misaligned with the gait-relevant rotation axes. Geodesic loss operates directly on $\text{tr}(\mathbf{R}^\top \hat{\mathbf{R}})$, a smooth function of rotation matrices, providing gradients that are proportional to the angular error without kinematic compounding.

What makes the GRU encoder work? The GRU hidden state at time t integrates all pose frames $\mathbf{p}_1, \dots, \mathbf{p}_t$. With geodesic supervision, the loss penalizes the *predicted future rotation* which is decoded from the hidden state. Backpropagation therefore shapes $\mathbf{h}_T$ to preserve the temporal structure of rotation changes—i.e., gait phase, turning direction, walking speed—which are precisely

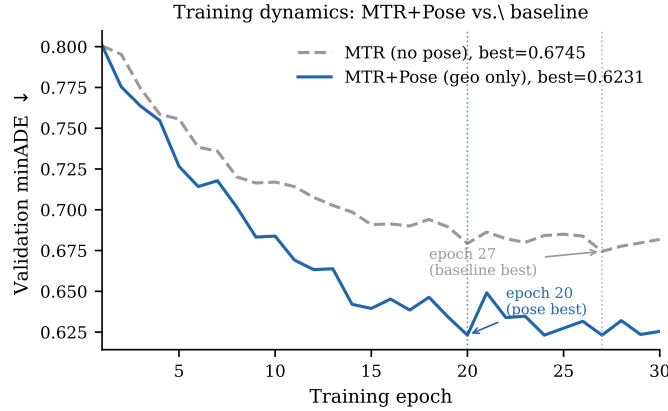


Figure 3: **Training dynamics.** Validation minADE over 30 epochs for MTR+POSE (geo only, blue) and the trajectory-only baseline (gray dashed). The pose-conditioned model converges earlier (best at epoch 20) with slightly higher epoch-to-epoch variance; the baseline plateaus more smoothly at a higher minADE.

the features useful for predicting where a pedestrian will walk next. Cross-attention without positional encoding produces a hidden state that is rotation-aware but *temporally unordered*: it knows what rotations occurred but not in what sequence, losing the gait dynamics that make pose useful for trajectory prediction.

The mechanism: past encoder, not future prediction. The null result of our future-pose ablation (Section 4.6) is strong evidence that the mechanism is *pose encoding*, not *pose prediction*. The GRU acts as a feature extractor for the MTR decoder, providing an orientation-aware agent representation that improves trajectory intention localization. The pose decoder (future prediction head) is an auxiliary task that supervises the encoder but does not itself contribute to trajectory accuracy.

6 LIMITATIONS

Dataset scale. Our experiments use 579 training scenes from Waymo, a small fraction of the full dataset. While relative comparisons are internally valid, we cannot claim that the 7.6% improvement transfers directly to a full-scale training regime.

SMPL pose availability. We use AMASS-derived pose sequences matched to Waymo pedestrian tracks, which requires a separate pose estimation step at test time. In deployment, online body pose estimation (e.g., from video) introduces estimation noise not present in our evaluation. The sensitivity of our method to noisy pose inputs is not evaluated here.

Cross-attention with positional encoding. Our H5 ablation tests cross-attention *without* temporal positional embeddings. Adding sinusoidal or learned PE could recover some or all of the GRU advantage. This variant is a natural next experiment.

No HD map features. The SMPL-paired subset lacks HD map polylines. We use zero-padded map features, which means our model does not benefit from map-aware pedestrian routing. Results on map-equipped data may differ.

Pose decoder quality. We do not evaluate MPJPE/geodesic error of the predicted future poses independently; our metric is trajectory prediction only. A joint evaluation of pose and trajectory quality would complete the picture.

7 CONCLUSION

We have presented MTR+POSE, a pose-conditioned extension of the Motion Transformer that incorporates past SMPL body poses via a GRU encoder with 6D rotation representations. Through a

systematic ablation of 13 runs on the Waymo Open Motion Dataset, we establish that: (1) geodesic supervision in $SO(3)$ is the critical ingredient, reducing minADE by 7.6% over the trajectory-only baseline, while joint-space MPJPE supervision yields only 2.5% improvement and degrades performance when combined with geodesic loss; (2) the GRU’s sequential temporal encoding is essential—cross-attention without positional encoding provides no improvement over the no-pose baseline; (3) the source of improvement is the past pose encoder, not future pose prediction quality.

These findings suggest that rotation-space supervision and temporal ordering are underappreciated design choices in pose-conditioned forecasting. We hope the systematic ablation methodology contributes to more principled design of multi-modal pedestrian prediction systems.

ACKNOWLEDGMENTS

The authors gratefully acknowledge access to GPU compute (RTX 4090) and the publicly available Waymo Open Motion Dataset.

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A ADDITIONAL EXPERIMENTAL DETAILS

A.1 ARCHITECTURE HYPERPARAMETERS

- GRU hidden size: 256
- Pose input dimension: 144 (24 joints $\times$ 6D rotation)
- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR encoder: 4-layer transformer, 256 hidden dim, 8 heads
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- LR schedule: Cosine decay, initial LR 1×10^{-3}
- Epochs: 30
- Gradient clipping: norm ≤ 1.0
- Dropout: none (small dataset)

A.2 DATASET CONSTRUCTION

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. Approximately 80% of validation pedestrians have a valid trajectory match; unmatched pedestrians are excluded from evaluation. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment.

A.3 LOSS WEIGHT STABILITY

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ± 1 boundary of the arccos argument; we clamp to $[-1 + \epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 FULL RESULTS TABLE

Table 4 reports all 13 experimental runs including invalid (zero future GT) baselines for completeness.

Table 4: Complete experimental results. Runs 001–002 are invalid (zero future GT) and excluded from analysis. Baseline for Δ is run 004.

Run	Tag	Encoder	Losses (w)	minADE	Δ
001	H1 full pose	GRU	all (1.0)	0.6114	<i>invalid</i>
002	H2 no pose	—	traj only	0.7724	<i>invalid</i>
003	H1_v2 full	GRU	all (0.1)	0.6532	−3.2%
004	H2_v2 baseline	—	traj only	0.6745	—
005	H1 w=0.05	GRU	all (0.05)	0.6557	−2.8%
006	H1 w=0.2	GRU	all (0.2)	0.6542	−3.0%
007	H1_v3 30fps	GRU	all (0.1), 30fps fut	0.6567	−2.6%
008	H3 mpjpe only	GRU	mpjpe+gmm (0.1)	0.6576	−2.5%
009	H3 gmm only	GRU	gmm (0.1)	0.6467	−4.1%
010	H3 geo only	GRU	geo+gmm (0.1)	0.6231	−7.6%
011	H3 geo w=0.05	GRU	geo+gmm (0.05)	0.6786	−1.5%
012	H3 geo w=0.2	GRU	geo+gmm (0.2)	0.6660	−1.3%
013	H5 cross-attn	XAttn	geo+gmm (0.1)	0.6765	+0.3%